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DIFFRACTIVE ANALYZER OF PATIENT TISSUE

Abstract

An x-ray diffractometer may perform 3D-analysis of collagen tissue of a patient (a human or another animal). The diffractometer includes an oblong housing that may be hinged and that contains an x-ray projector and an x-ray receiver. The analyzed tissue, such as the external ear and skin of a patient, is accommodated between the x-ray projector and the x-ray receiver. The x-ray projector directs an x-ray micro-beam at the patient's tissue. The receiver contains a movable two-dimensional x-ray detector that detects the x-ray micro-beam passed through the analyzed tissue and detects x-rays scattered or diffracted by the analyzed tissue.

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Background/Summary

REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. patent application Ser. No. 18/137,342, filed on Apr. 20, 2023, which claims priority to U.S. Provisional Patent Application No. 63/333,079, filed Apr. 20, 2022, all of which are hereby incorporated by reference in their entirety.

BACKGROUND

[0002] Early diagnosis of a malignancy directly correlates with successful treatment of the malignancy. Unfortunately, no noninvasive, cost-effective diagnostic tests are readily available for many malignancies, which results in patients often presenting too late for effective treatment.

[0003] Fiber diffraction patterns of skin or fingernails, using x-ray sources, has been proposed as a biometric diagnostic method for detecting neoplastic disorders including but not limited to melanoma and breast, colon, and prostate cancers. Veronica J. James, “Fiber diffraction of skin and nails provides an accurate diagnosis of malignancies”, Int. J. Cancer: 125, 133-138 (2009), suggests that with further development, procedures using small angle x-ray beam lines at synchrotrons may provide a confirmatory or diagnostic tests that could be conducted on a regular basis in local radiology facilities. In such tests, samples of human and animal hair, nails, and skin could be used for “indirect” detection and diagnosis of various diseases.

[0004] The effect of different endogenous and exogenous factors on the molecular and nano structural ordering of human hair has been studied using x-ray fluorescence and diffraction from synchrotron radiation. The diffraction hair causes may be attributed to two fibrillar systems of tissue: the intermediate keratin filaments of the cytoskeleton of the hair and the proteoglycan fibrils of the extracellular matrix of the hair. The effect that personal hygiene products and medicines have on the structural transformation and elemental composition of hair may be investigated. Proteoglycans are considered as universal components of a matrix that ensure the structural homeostasis of biological tissue subjected to endogenous and exogenous effects. Hair may be a promising biological material for solving applied problems when used as a diagnostic material for the widescale monitoring of environmental and public health risks.

[0005] Mark R. Goldstein, Luca Mascitelli, “Might tumor secreted cathepsin proteases leave specific molecular signals in skin, hair and nails years before a cancer becomes clinically apparent?”, Medical Hypotheses 103 (2017) 62-63, suggests that X-ray fiber diffraction analysis (FDA) of the fibrous macromolecules in hair, nails and skin may allow diagnosis of various cancers, at sites remote from the cancer and years before the cancer becomes clinically apparent. Currently, this technology is not widely accepted because of reproducibility issues and a lack of an explanation as to how a clinically unapparent tumor can leave molecular “signatures” at remote sites. However, some evidence suggests that tumor-specific cathepsins (lysosomal proteases) circulate systemically long before a cancer is clinically apparent. One possible mechanism that could leave a signature for FDA is that cathepsins, by virtue of their proteolytic activity, impart molecular changes in tissues remote from the primary tumor. FDA of hair, nails, and skin might

detect these subtle molecular changes, which may be specific for various tumors.
[0006] Some publications have indicated that the structure of collagen changes in patients (animals and humans) suffering from severe oncological diseases.
[0007] Synchrotron based x-ray diffraction experiments may also be effective in the study of mammalian connective tissues and related diseases. Rama Sashank Madhurapantula and Joseph P. R. O. Orgel, “X-Ray Diffraction Detects D-Periodic Location of Native Collagen Crosslinks In Situ and Those Resulting from Non-Enzymatic Glycation”, http://dx.doi.org/10.5772/intechopen.71022 describe observing changes in the structure of Extra-Cellular Matrix (ECM), induced in an ex-vivo tissue based on a model of the disease process underlying diabetes. Pathological changes to the structure and organization of the fibrillar collagens within the ECM, such as the formation of nonenzymatic crosslinks in diabetes and normal aging, have been shown to play an important role in the progression of such maladies. However, without direct, quantified, and specific knowledge of where in the molecular packing these changes occur, development of therapeutic interventions has been impeded. In vivo, the result of non-enzymatic glycosylation e.g., glycation, is the formation of sugar-mediated crosslinks, aka advanced glycation end-products (AGEs), within the native D-periodic structure of type I collagen. The locations for the formation of these crosslinks have been inferred from indirect or comparatively low-resolution data under conditions likely to induce experimental artifacts. Sashank et al. indicate x-ray diffraction derived data, collected from whole hydrated and intact isomorphically derived tendons, that indicate the location of both native (existing) and AGE crosslinks in situ of D-periodic fibrillar collagen.
[0008] Biophysical properties of an extracellular matrix (ECM), such as matrix stiffness, viscoelasticity, and matrix fibrous structure, are emerging as important factors that regulate progression of fibrosis and other chronic diseases. See Wenyu Kong et al, “Collagen crosslinking: effect on structure, mechanics and fibrosis progression”, 2021 Biomedical Materials 16 062005. The biophysical properties of the ECM can be rapidly and profoundly regulated by crosslinking reactions in enzymatic or non-enzymatic manners, which further alter the cellular responses and drive disease progression. In-depth understandings of crosslinking reactions may help reveal the underlying mechanisms of fibrosis progression and put forward new therapeutic targets, whereas related reviews are still devoid. Kong et al. focus on the main crosslinking mechanisms that commonly exist in many chronic diseases (e.g., fibrosis, cancer, osteoarthritis) and summarize current understandings including the biochemical reaction, the effect on ECM properties, the influence on cellular behaviors, and related studies in disease model establishment.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0009] FIG. **1** is a block diagram illustrating an example of an x-ray diffractometer in accordance with an example of the present disclosure.
[0010] FIG. **2** schematically shows a perspective view of an x-ray diffractometer system according to an example of the present disclosure.
[0011] FIG. **3** schematically shows the perspective view of an x-ray diffractometer system according to another example of the present disclosure.
DETAILED DESCRIPTION
[0012] A diffractometer in accordance with one example disclosed herein may be used to study the phase composition of collagen tissue and disturbances of periodic structure in collagen tissue. The phase composition and disturbances, which illness may cause, may be identified to develop diagnostic criteria based on diffractometer measurement data. The diffractometer may be particularly adapted to analyze a patient's external ear and skin, which generally contains multiple types of collagen, e.g., collagen I and collagen II.

[0013] An x-ray system in accordance with another example of the present disclosure may provide x-ray 3D-analysis of an external ear and skin collagen tissue of a human or other animal. The x-ray system may be used to detect matrix related diseases, including, but not limited to, inflammation, melanoma, breast, colon, and prostate cancer and used to the study of changes in tissue structure of these foci. The x-ray system may be suitable for human and veterinary medicine and may provide easily accessible, non-invasive, cost-effective diagnostic tests at early stages of diseases.
[0014] A diffractometer in accordance with another example of the present disclosure may provide x-ray 3D-analysis of external ear and skin collagen tissue of a patient, i.e., a human or other animal. A specific example of a diffractometer may include: an oblong housing defining a recess where the external ear and skin of a patient may be placed. X-ray devices may be located in the housing on opposite sides of the recess. One x-ray device in the diffractometer may be an x-ray projector that produces and directs a primary incident micro-beam of x-rays at analyzed tissue, e.g., the external ear and skin of a patient, and another x-ray device may be a receiver system including a movable two-dimensional pixel detector capable of detecting the transmitted micro-beam of x-rays passed through the analyzed tissue and detecting some or all x-rays that the analyzed tissue diffracts from the incident micro-beam beam.
[0015] X-ray systems or diffractometers in accordance with some examples disclosed herein may provide 3D analysis of the molecular or nano-scale structure of living tissue, particularly of the external ear and skin of a human or other animal patient and more particularly of collagen tissue in the external ear or skin of the patient.
[0016] FIG. **1** shows a block diagram of a diffractometer **100** in accordance with an example of the present disclosure. Diffractometer **100** includes an x-ray projector **110** and an x-ray detector **120**. X-ray projector **110** directs an x-ray beam **142** into a tissue holding area **140**, and x-ray detector **120** is positioned to detect an x-ray scattering or diffraction pattern resulting when x-ray beam **142** from x-beam projector **110** interact with tissue of a patient, e.g., an external ear or skin **144** of a human or other animal being examined.
[0017] X-ray projector **110** in the illustrated example includes an x-ray source **112** and an x-ray beam shaper or conditioner **114**. X-ray source **112** may include, for example, an x-ray tube or x-ray laser. X-ray beam conditioner **114** may include one or more collimating systems **115**, focusing systems **116**, or filter systems **117**. For example, collimating systems **115** may include a Kratki or Montel mirror collimator. Focusing systems **116** may include grazing-incidence mirrors that are slightly curved to focus x-ray beam **142**. Filter systems **117** of x-ray beam conditioner/shaper **114** may include, for example, at least one monochromator. In one specific example, x-ray source **112** includes a radiation source operating in continuous mode, e.g., x-ray tube, to produce monochromatic x-rays and x-ray beam conditioner **114** includes a collimating mirror and aperture that form or shape the x-ray micro-beam **112**. Alternatively, x-ray beam conditioner **114** may employ any other systems capable of controlling the width, collimation, or other characteristics of x-ray beam **142**.
[0018] In operation, beam conditioner **114** receives x-rays from x-ray source **112** and produces an x-ray beam **142** having desired intensity, wavelength, and beam profile. X-ray beam **142** may, for example, be is a micro-beam having a FWHM of about 200 microns or less, a wavelength of 1.60 Å to 0.3 Å, and an intensity on the order of about 10.sup.12 photons/sec/mm.sup.2.
[0019] X-ray beam **142** from x-ray beam projector **110** passes into area **140** where patient tissue such as an external ear or skin **144** of the patient (human or other animal) may be presented. X-ray beam **142** enters and interacts with external ear or skin **144**, where x-rays scatter or diffract. As a result, both transmitted, i.e., unscattered, x-rays **146** and scattered x-rays **148** enter x-ray detector **120**.
[0020] X-ray detector **120** in the illustrated example includes a detector array **122** or other system capable of detecting a distribution of x-rays. Detector array **122** may be a two-dimensional array of pixel sensors, with each pixel sensor being a photodiode or transistor that is sensitive to x-rays.

Detector array **122** is mounted on a detector mounting system or stage **124** that is operable to move detector array. In the illustrated example, detector mounting system **124** includes an actuator or detector position adjuster **126** capable of mechanically moving pixel detector **122** toward or away from tissue holding area **140** along the propagation direction of incident x-ray beam **142** and transmitted x-ray beam **146**. Detector position adjuster **124** may, for example, include a drive train **126** of a type used to adjust the height of optics assemblies in photocopiers.

[0021] Movement of pixel detector array **122** relative toward or away from the analyzed tissue, e.g., the external ear and skin **144** of the patient, can change the measurement resolution and the measurement range of radiation **148** that the analyzed tissue scatters. For instance, moving detector array **122** further from the analyzed tissue increases the area of detector array **122** exposed to radiation **148** scattered over small angles and thereby improves the ability of x-ray detector **120** and array **122** to resolve a small angle scattering pattern. On the other hand, moving detector array **122** closer to the analyzed tissue **144** enables radiation **148** that the analyzed tissue **144** scatters or diffracts to larger angles to strike detector array **122**, allowing x-ray detector **120** and array **122** to measure radiation **148** scattered at larger angles, e.g., close to 90° if detector is sufficiently close to analyzed tissue **144**. Diffractometer **100** may thus position detector array **122** to measure x-ray scattering at a desired resolution or in a desired range of angles. For examples, movement along the beam direction can position detector array **122** for high resolution diffraction measurements over a small range of angles (e.g., angles less than 1°) or for diffraction measurements over a large range of angles (e.g., angles approaching 90°).

[0022] Pixels in detector array **122** generally need to be sensitive enough to detect or measure scattered x-rays, which are generally much less intense than transmitted x-rays **146**. To extend the operational life of detector array **122**, detector mount **124** for detector array **122** may further include a mechanism **128** that excites transverse vibrations of detector array **122**. The vibrations move detector array **122** so that transmitted x-rays **146** are not constantly striking the same pixel or pixels in detector array, but instead, strike different pixels at different times. By exciting vibrations of detector array **122** that are transverse relative to transmitted x-ray beam **146** and larger in amplitude than the width of transmitted x-ray beam **146**, mechanism **128** may prevent the high intensity of transmitted x-ray beam **146** from quickly burning out the central pixel or pixels, i.e., the pixels of detector array **122** that measure the transmitted x-ray beam **146** and identify the center of an x-ray diffraction pattern. X-ray diffraction pattern measurements, e.g., frames captured by detector array **122**, at different times during a transverse vibration of detector array **122** can be electronically or digitally centered and combined or calibrated to account for the difference in the transverse position detector array **122**.

[0023] Different examples of diffractometer **100** may use different types of transverse vibrations of detector array **122**. In one example disclosed herein, diffractometer **100** drives mechanism to excite a longitudinal surface acoustic wave causing transverse vibrations of detector array **122**. The vibrations may include linear fluctuations only in the equatorial direction (e.g., perpendicular to the page of FIG. **1**), linear fluctuations only in the meridional direction (e.g., vertical in FIG. **1**), or two-dimensional fluctuations in the equatorial and meridional directions simultaneously.

[0024] Mechanism **128** in one example contains piezoelectric actuators and uses the piezoelectric effect to drive the transverse vibration or oscillations of detector array **122**. The amplitude of the transverse vibrations of detector array **122** may be at least as large as the pitch of pixels in detector array **122** or the FWHM width of transmitted beam **146**. In still another aspect disclosed herein, the frequency of the transverse vibrations of detector array **122** may be less than or equal to a frame rate at which the two-dimensional pixel detector acquired measurements. In general, the frequency of oscillations may be several, e.g., 4 to 6, orders of magnitude less than the frequency of frame capture. A readout chip **125** for detector array **122** can control the speed and timing of transverse movements of detector array **122** or a frame rate at which detector array **122** captures measurements to avoid smearing of the x-ray pattern measurements that detector array **122** captures

in respective frames.
[0025] In yet another aspect disclosed herein, readout chip **125** implements a known operating mode, sometimes referred to as Charge Summing Mode (CSM), with the purpose of eliminating charge-shared events. More particularly, x-ray photons striking pixel sensors in an array may cause charge accumulation or other effects on neighboring pixels. In one case, readout chip **124** can sum signals from neighboring pixels at a series of "summing nodes" and assign each hit to the node or pixel with the highest signal, to thereby compensate for charge-sharing and similar effects.
[0026] In accordance with another aspect of the present disclosure, diffractometer **100** does not employ a beam stop. Avoiding use of a beam stop near detector array **122** and particularly avoiding need of a beam stop in front of detector array **122** avoids parasitic scattering or backscattering of x-rays from the beam stop and may reduce noise and provide more accurate measurement or detection of scattering of x-ray micro-beam **142** of x-rays from analyzed tissue **144**.
[0027] In another aspect disclosed herein, diffractometer **100** may include an oblong housing **130** that is hinged. In the example of FIG. **1**, housing **130** includes a source portion **132**, a receiver portion **134**, and a swivel joint or hinge **136** that connects source portion **132** to receiver portion **134** and allows folding of housing **130**, e.g., for transportation of diffractometer **100**. Source portion **132** of housing **130** may contain or provide a mounting for x-ray beam projector **110** on one side of hinge **136**, and receiver portion **134** may attach to or contain mounting **124** for detector array **122** on the other side of hinge **136**. Interiors of source portion **132** and receiver portion **134** may be at a vacuum pressure or filled with an inert gas, e.g., neon or helium, and source portion **132** or receiver portion **134** may have a wall or window **133** and **135** that faces tissue holding area **140** and that is transparent to x-rays. In particular, x-ray projector **110** may be sealed in source portion **134** (or a protection container) that is vacuumed or filled with an inert gas, and detector array **122** may similarly be in receiver portion **134** or another protection container that is vacuumed or filled with an inert gas.
[0028] In still another example disclosed herein, one or more of x-ray beam projector **110**, source portion **132**, and receiver portion **134** may be equipped with telescoping structures **150** that can extend to (and retract from) contact with the surface of analyzed tissue **144** to hold or fix analyzed tissue **144** during x-ray diffraction analysis. Each telescoping structure **150** may surround the optical paths of incident x-ray beam **142** or transmitted x-rays **146** and scattered x-rays **148** and may contain a vacuum or an inert gas to reduce any scattering or attenuation of scattered x-rays **148** between analyzed tissue **144** and detector array **122**.
[0029] In yet another example disclosed herein, each telescoping structure **150** may be equipped with a window **152**, e.g., a beryllium window, that is transparent to x-rays. Windows **152** may seal off the vacuum areas inside telescoping structures **150** from the non-vacuum area **140** for patient tissue **144**. In yet another example, telescoping structures **150** further include a contact pressure sensor **154** that senses a force or compression that extension of the telescoping structures **150** applies to the external ear and skin or other patient tissue **144** so that a control system of diffractometer **100** can regulate the force that telescoping structures **150** apply to the patient.
[0030] In yet another example, diffractometer **100** may further include a visible light laser **118**. Visible light laser **118** may be aligned with or part of x-ray beam projector **110**, so that a visible beam **162** from laser **118** may be used to place a visible spot on analyzed patient tissue **144** for pointing of the diffractometric x-ray beam **142** at a selected point on analyzed tissue **144**.
[0031] In another example, a mobile version of diffractometer **100** has a small size, e.g., less than about 1.5 m in length when fully extended. The mobile version of diffractometer **100** may employ hinge **136** to fold diffractometer **100** so that source portion **132** and receiver portion **134** are adjacent to each other and the length of diffractometer **100** is shortened bay about one half, e.g., less than 75 cm. The mobile version of diffractometer **100** may be equipped with an autonomous power source **160**, which may be based on electric accumulators or batteries. In another example, diffractometer **100** is designed to be installed on vehicle such as an automobile, and power source

160 for diffractometer **100** is an automotive electric generator, battery, or other automotive electrical system.

[0032] FIGS. **2** and **3** are schematic representations of examples of alternative diffractometer systems in accordance with specific examples, and elements illustrated in the drawings are not necessarily shown to scale. Rather, the various elements are represented in FIGS. **2** and **3** to illustrate the functions and general purposes of the elements. Any connection or coupling between functional blocks, devices, components, or other physical or functional units shown in the drawings or described herein may also be implemented by an indirect connection or coupling. A coupling between components may also be established over a wireless connection. Functional blocks may be implemented in hardware, firmware, software, or a combination thereof.

[0033] FIG. **2** schematically shows a perspective view of an x-ray diffractometer system **200** for x-ray 3D-analysis of collagen tissue such as the external ear and skin of a patient (not shown). Diffractometer system **200** includes a diffractometer **230** having an x-ray source portion **232**, an x-ray receiver portion **234**, and a hinge or swivel joint **236** connecting x-ray source portion **232** and x-ray receiver portion **234**. Source portion **232** and receiver portion **234** are shaped and connected (via joint **236**) to define a recess **240** that may surround the external ear or other tissue of a patient being examined. X-ray source portion **232** contains a source of an x-ray beam that directs an x-ray micro-beam into recess **240**, and swivel joint **236** may be used to position a detector in x-ray receiver portion **234** to receive and measure x-rays that are transmitted through or scattered by the patient's tissue. For example, swivel joint **236** may position x-ray receiver portion **234** straight across from x-ray source portion **232**, so that a center point of a detector in x-ray receiver portion **234** is aligned with the x-ray beam from x-ray source portion **232**. Swivel joint **236** further permits folding of x-ray source portion **232** and x-ray receiver portion **234** into a more compact configuration, e.g., for transportation.

[0034] X-ray source portion **232** and x-ray receiver portion **234** may be equipped with telescoping structures **250** that may be retracted at the beginning of an x-ray diffraction examination process and then extended to fix a patient's external ear and skin in place for the x-ray diffraction examination. X-ray source portion **232** and x-ray receiver portion **234** may include separate containers, each holding a vacuum around the x-ray source or receiver components, and one or more sealed beryllium walls or windows **252** may be located at one or both ends of each telescoping structure **250**. X-ray receiver portion **234** may contain a movable two-dimensional pixel detector designed to detect the transmitted micro-beam of x-ray passed through the analyzed tissue as well as detect part or all x-rays that are diffracted or otherwise scattered by the analyzed tissue. X-ray receiver portion **234** may further include a mechanism for moving the movable two-dimensional pixel detector along the direction of the incident micro-beam of x-ray, and the two-dimensional pixel detector may provide approximately the same relative resolution of diffraction measurements at small angles (less than about 1°) and large angles (up to about 90°). The mechanisms moving the sliding two-dimensional detector, by way of example and not by way of limitation, may employ a ball-screw motion transmission, which transmits the rotational movement from an electrical motor to the translational movement of the two-dimension detector. X-ray receiver portion **234** may also include one or more devices designed for excitation of vibrations of the two-dimensional pixel detector that are transverse relative to the incident x-ray beam. In general, the mechanism providing transverse movement or vibration of the detector array may operate independently of the mechanism that moves the detector array along the beam direction.

[0035] Diffractometer **230** attaches to a mounting structure such as a tripod **290** with collapsible (demountable) connecting device, and tripod **290** may include motorized or manual mechanism that permit positioning and orienting diffractometer **230** to accommodate a location on a patient of the tissue to be examined. Such movement systems may include adjustment systems **292** and **293** for the height and separation of the diffractometer **230** from the base of tripod **290** and adjustment systems **294**, **296**, and **298** for control of the yaw, pitch, and roll of diffractometer **230**.

[0036] A computer workstation **280**, which may be a conventional general-purpose computer with hardware such as one or more microprocess with associated memory and I/O interfaces, may execute software or firmware to control drive systems of tripod **290** to control the position and orientation of diffractometer **230**. Computer workstation **280** may also receive and process measurement from diffractometer **230**, conduct diffractometric structural analysis using the measurements received, processes or identify one or more diffraction patterns that the structural analysis found, and store and display the obtained data. Computer workstation **280** may exchange data and control signals with diffractometer **230** or tripod **290** over a connection **282**, which may, for example, be a wired or wireless communication channel or network device.
[0037] Computer workstation **280** in some diffractometer systems may be configured and connected to control one or multiple x-ray devices, e.g., x-ray sources and detectors, control drive mechanisms and motors, e.g., for detector position or orientation adjustment, and may acquire, store, and process measurement data from multiple x-ray detectors. Computer workstation **280** may process, store and display data or results from the 3-D diffractometric structural analysis, e.g., from calculation of parameters of the three-dimensional reciprocal lattice of the studied collagen tissue.
[0038] FIG. **3** shows an example of an x-ray diffractometer system **300** including an x-ray diffractometer **330** held in a wall or a ceiling mount **310**, which may be installed in a laboratory or examination room. Other than its mounting structure, diffractometer **330** may be substantially identical to diffractometer **200** of FIG. **2**, and diffractometer system **300** may contain the same or similar components to those described above with reference to FIG. **2**. Wall or ceiling mount **310** includes a telescoping arm **310** extending between a wall or ceiling mounting structure **314** and a ball joint connection structure **316** for diffractometer **330**. Mount **310** may allow independent movement of arm **312** along an actuation direction of wall or ceiling mounting **314**, rotation of arm **312** about a length axis of arm **312**, extension and retraction of the length of arm **312**, and pitch, roll, and yaw rotations of diffractometer on ball joint **316** for control of the position and orientation of diffractometer **330** in three dimensions. Mount **310** may further include mechanisms and motors designed to move telescoping arm **312** and mounting structures **314** and **316** along their respective degrees of freedom of motion. In yet another aspect disclosed herein, movement of diffractometer **330** may be realized by using appropriate mechanisms such as a ball-screw motion transmission or a combination of a feed screw shaft that is driven to rotate by a position-controllable electric motor and a slide member that is held in thread-engagement with the feed screw shaft as well as using a step motor. In another example, a rotation of diffractometer **330** may be realized by using appropriate rotary drive mechanisms such as a tangent bar type drive mechanism where the drive source is preferably a position-controllable electric motor such as a pulse motor or a servomotor.
[0039] Although aspects of the present disclosure have been described in detail with reference to certain implementations or examples, persons possessing ordinary skill in the art to which this disclosure pertains will appreciate that various modifications and enhancements may be made without departing from the spirit and scope of the claims that follow. Any feature, whether preferred or not may be combined with any other feature whether preferred or not. Alternatives to the examples described herein can be employed in practicing the invention as defined in the following claims. It is intended that the following claims define the scope of the invention and that methods and structures within the scope of these claims and their equivalents be covered thereby. The appended claims are not to be interpreted as including means-plus-function limitations, unless such a limitation is explicitly recited in a given claim using the phrase “means for.”

Claims

1. A diffractometer system comprising: an x-ray source including an x-ray projector that projects an x-ray beam; an x-ray receiver including an array of x-ray sensors positioned to detect x-rays from the x-ray beam, wherein patient tissue to be analyzed can be accommodated between the x-ray source and the x-ray receiver; and one or more telescoping structures on at least one of the x-ray source and the x-ray receiver, each telescoping structure extending to and fixing the patient tissue in place during an x-ray examination process.

2. The diffractometer system of claim 1, further comprising a computer system connected to the x-ray projector and the x-ray receiver, wherein the computer system is configured to control the x-ray projector and the x-ray receiver, process, store, and display data received from the x-ray receiver, perform 3-D diffractometric structural analysis, and calculate parameters of a three-dimensional reciprocal lattice of collagen in the patient tissue.

3. The diffractometer system of claim 1, wherein the x-ray projector comprises a radiation source and a beam conditioner forming the x-ray beam, the radiation source operating in continuous mode, and the beam conditioner including at least one monochromator, an x-ray collimating device, and an x-ray focusing device.

4. The diffractometer system of claim 1, wherein the patient tissue includes an external ear of a patient, and each of the telescoping structures presses against a skin surface of the external ear to hold the patient tissue in place for the x-ray examination process.

5. The diffractometer system of claim 4, where at least one of the telescoping structures further comprises a pressure sensor, the pressure sensor measuring force or compression applied to the external ear.

6. The diffractometer system of claim 1, wherein the x-ray source and the x-ray receiver are sealed to hold a vacuum or an inert gas and are equipped with windows through which the x-ray projector projects the x-ray beam and the x-ray receiver receives the x-rays.

7. The diffractometer system of claim 1, wherein the array of x-ray sensors is in a protection container that is vacuumed or filled with an inert gas.

8. The diffractometer system of claim 1, further comprising a swivel joint that connects the x-ray source to the x-ray receiver and permits rotation of the x-ray receiver relative to the x-ray source.

9. The diffractometer system of claim 1, wherein the x-ray receiver further comprises: a first mechanism connected to move the array of x-ray sensors in a first direction toward or away from the x-ray source; and a second mechanism coupled to excite vibrations of the array of x-ray sensors in a second direction that is transverse to the first direction.

10. The diffractometer system of claim 9, where the second mechanism is coupled to excite one or more types of transverse vibrations selected from a group consisting of fluctuations only in an equatorial direction perpendicular to the first direction, fluctuations in a meridional direction perpendicular to the first direction and to the equatorial direction, and fluctuations in both the equatorial and meridional directions simultaneously.

11. The diffractometer system of claim 1, wherein the x-ray projector comprises a radiation source selected from a group consisting of an x-ray tube and an x-ray laser.

12. The diffractometer system of claim 1, further comprising a tripod including a motorized positioning mechanism coupled to move and orient the x-ray source and the x-ray receiver.

13. The diffractometer system of claim 12, wherein the motorized positioning mechanism comprises at least one of a ball-screw motion transmission and a combination of a feed screw shaft that is driven to rotate by a position-controllable electric motor and a slide member that is held in thread-engagement with the feed screw shaft as well as using a step motor.

14. The diffractometer system of claim 1, wherein the x-ray projector further comprises a Kratki or Montel mirror collimator.

15. The diffractometer system of claim 1, further comprising a visible-light laser providing a light beam that is parallel to the x-ray beam.

16. The diffractometer system of claim 1, further comprising an autonomous power source based on electric accumulators and batteries.

17. The diffractometer system of claim 16, wherein the autonomous power source is an automotive electrical system.

18. The diffractometer system of claim 1, further comprising: a telescoping arm attached to the x-ray source and the x-ray receiver and to a wall or a ceiling of a room; and mechanisms and motors coupled to move the telescoping arm.